

Data Intake Processing and Verification Report

Background Information

- **Original Dataset Name:** The ECCO-Darwin Data-Assimilative Global Ocean Biogeochemistry Model: Estimates of Seasonal to Multidecadal Surface Ocean $p\text{CO}_2$ and Air-Sea CO_2 Flux
- **GHG Center Dataset Title:** ECCO-Darwin Modeled CO_2 -Ocean Flux
 - Title prior to September 2025: Air-Sea CO_2 Flux, ECCO-Darwin Model v5
- **Dataset Provider:** NASA
- **Date Obtained:** August 2023
- **Location Obtained From:** Direct data delivery via private Google drive
 - Note: source data in binary format available from:
https://data.nas.nasa.gov/ecco/llc_270/ecco_darwin_v5/output/monthly/CO2_flux/
- **Data Location in GHG Center:** eccodarwin-co2flux-monthgrid-v5
- **Data POC(s):** Dr. Dustin Carroll, Dr. Kevin Bowman
- **Dataset File Type(s):** NetCDF
- **Projection (if different from WGS84):** NA

Data Transfer Confirmation

An SHA-256 checksum is used to detect high-level errors within data transmissions. Results from individual checksum file comparisons of pre-transfer and post-transfer shows all files were transferred successfully and no individual files had any transfer issues.

Data Intake Process

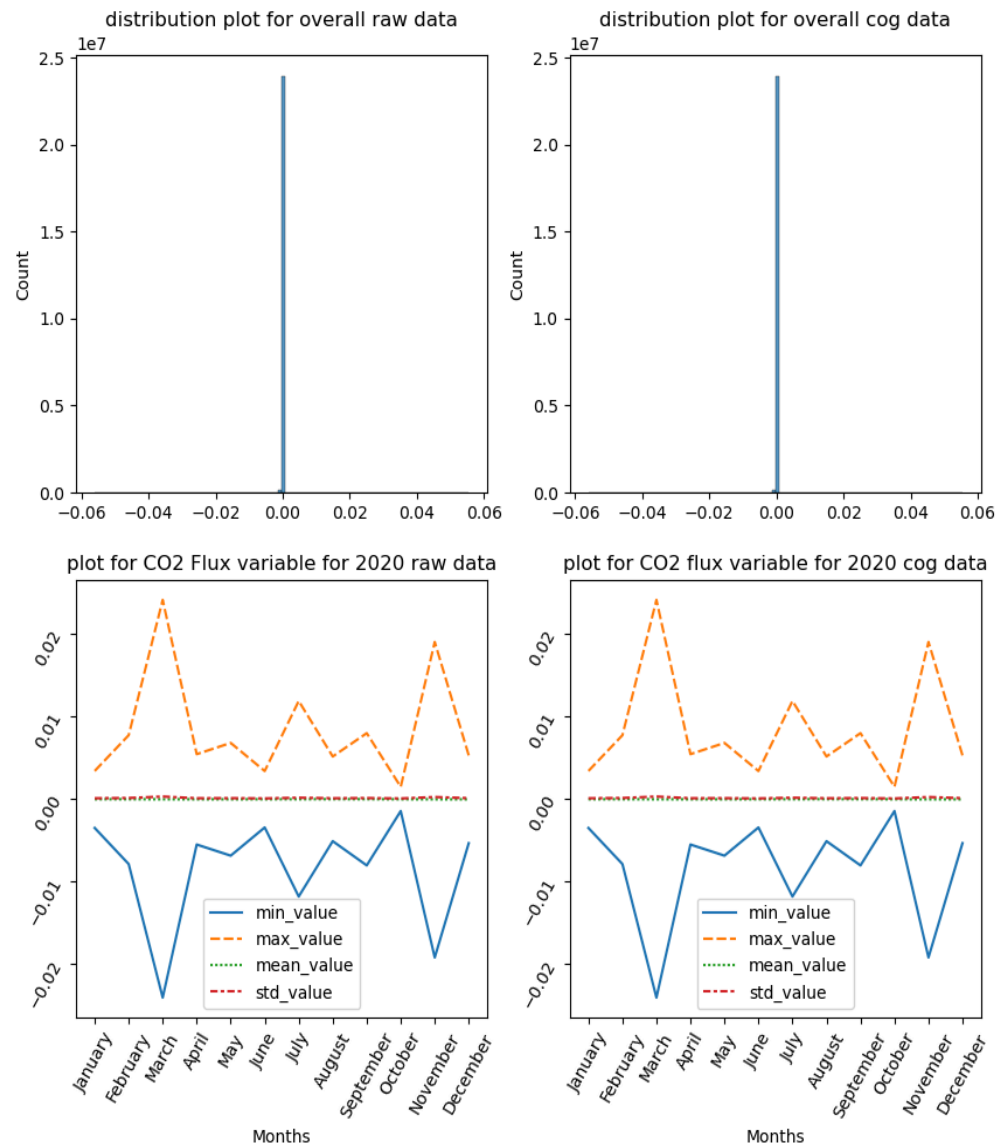
- https://us-ghg-center.github.io/ghgc-docs/data_workflow/eccodarwin-co2flux-monthgrid-v5_Data_Flow.html

Overall Dataset Statistics

- Number of data files read: 36 NetCDF (original data) and 36 COGs (transformed cloud optimized GeoTIFFs)
- Statistics across all 36 files:

	Minimum (mmol m^2/s)	Maximum (mmol m^2/s)	Mean (mmol m^2/s)	Standard Deviation
Original Data	-0.05605	0.05544	-0.0000262	0.00019
Transformed Data	-0.05605	0.05544	-0.0000262	0.00019

- Distribution of values in $\text{mmol m}^2/\text{s}$ across files:



- Statistics for November 2022:

	Minimum ($\text{mmol m}^2/\text{s}$)	Maximum ($\text{mmol m}^2/\text{s}$)	Mean ($\text{mmol m}^2/\text{s}$)	Standard Deviation
Original Data	-0.0310	0.0307	-0.000022	0.000377
Transformed Data	-0.0310	0.0307	-0.000022	0.000377

- All values are in expected range

Summary

- We are confident that the transformation and display of data in the US GHG Center accurately represents the source data
- [Link](#) to the transformation code
- [Link](#) to the dataset Tutorial
- [Link](#) to the US GHG Center data catalog page

Report Completed on: 9/27/2023, updated on 8/18/25 to reflect updated dataset title

MSFC POC for questions: [Jeanné le Roux](#), [Siddharth Chaudhary](#)